

Lakshmi Shivani K

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Profile

Cybersecurity engineering enthusiast with hands-on experience in security automation, API security testing, OSINT-based attack surface analysis, and Python-based security tool development. Skilled in vulnerability assessment, network security concepts, and cybersecurity automation.

Education

M.Tech in Cyber Security

Nov 2025 – Nov 2027

REVA Academy for Corporate Excellence (RACE)
REVA University, Bengaluru, India

B.E. in CSE (IoT & Cyber Security including Blockchain Technology)

Dec 2021 – Jun 2025

SEA College of Engineering and Technology, Bengaluru, India
CGPA 7.9

Skills

Technical Skills

Programming: Python, Embedded C, Java

Security & Analysis: Networking Fundamentals, API Security, Security Automation, Authentication & Access Control, OSINT Attack Surface Analysis, Threat Intelligence, Digital Footprint Analysis, Threat Detection & Basic Threat Hunting, Incident Investigation & Basic Incident Triage, Vulnerability Assessment, OWASP API Top 10

Tools

Nmap, Nikto, Postman, Wazuh, Wireshark, Wappalyzer, Arpspoof, Linux & Shell Basics, Git, Power BI, AWS Console (Lambda, API Gateway, DynamoDB)

Soft Skills

Analytical Thinking, Problem Solving, Attention to Detail, Communication, Team Collaboration, Technical Documentation

Internship

Machine Learning Intern

Feb 2025 – May 2025

Psitron Technologies Pvt Ltd

- Contributed to Generative AI projects focused on building scalable solutions.
- Evaluated LLMs to enhance response accuracy and system performance.
- Leveraged AWS services to design and manage backend infrastructure.
- Developed serverless applications using AWS Lambda and API Gateway.

Projects

Adversary-Centric Digital Footprint Analyser

- Developed a Python-based OSINT framework for analysing organisational attack surface exposure

- Automated collection of subdomains, open ports, leaked credentials, technology stack details, and email patterns
- Designed a weighted risk scoring model to generate composite attack surface risk scores
- Evaluated multiple university domains to identify exposed assets and potential security risks
- Generated structured HTML security assessment reports with prioritised findings and remediation guidance

ARP Spoofing Detection Tool

- Developed a network security tool using Python and Scapy to detect ARP spoofing (MITM attacks).
- Monitored IP-MAC inconsistencies to identify suspicious network activity in real time.
- Performed packet analysis to detect abnormal ARP responses and potential threats.
- Generated alerts for detected anomalies and suspicious behavior.
- Validated detection accuracy by simulating attacks using tools like arpspoof.

AI Command Line Assistant for Cybersecurity Automation

- Built a Python-based automation tool using subprocess and regex for executing security tasks.
- Automated reconnaissance operations including Nmap scanning, WHOIS lookup, ping, and traceroute.
- Implemented dynamic command parsing for efficient and intelligent tool execution.
- Integrated SHA-256 hashing for file integrity verification and basic incident analysis.
- Developed a user-friendly interface using Gradio to simplify security operations.

API Security Assessment Engine (OWASP Top 10)

- Conducted API security testing using Postman and Python scripts to identify OWASP vulnerabilities.
- Simulated real-world attack scenarios to detect authentication and access control issues.
- Identified security flaws such as misconfigurations and improper input validation.
- Generated structured reports with risk analysis and remediation recommendations.
- Designed a modular approach to support testing across different REST APIs.

Certifications

- Cyber Security Fundamentals
- Networking Basics
- Python 101 for Data Science